Exercise 15B

Solution 1: Here, $r = 28/2 = 14$ cm; $h = 40$ cm

Curved surface area of the cylinder $= 2\pi rh$

$$= 2 \times \frac{22}{7} \times 14 \times 40 \text{ cm}^2$$
$$= 2 \times 22 \times 2 \times 40 \text{ cm}^2$$
$$= 3520 \text{ cm}^2$$

Total surface area of the cylinder $= 2\pi rh + 2\pi r^2$

$$= \left(3520 + 2 \times \frac{22}{7} \times (14)^2\right) \text{ cm}^2$$
$$= \left(3520 + 2 \times \frac{22}{7} \times 14 \times 14\right) \text{ cm}^2$$
$$= (3520 + 2 \times 22 \times 2 \times 14) \text{ cm}^2$$
$$= (3520 + 1232)$$
$$= 4752 \text{ cm}^2$$

$\therefore$ Volume of the cylinder $= \pi r^2 h$

$$= \frac{22}{7} \times (14)^2 \times 40 \text{ cm}^3$$
$$= \frac{22}{7} \times 14 \times 14 \times 40 \text{ cm}^3$$
$$= 22 \times 14 \times 2 \times 40 \text{ cm}^3$$
$$= 24640 \text{ cm}3$$

Solution 2:

Radius of the bowl, $r = \frac{7}{2}$ cm

Height of soup in the bowl, $h = 4$ cm

$\therefore$ Volume of soup in one bowl $= \pi r^2 h$

$$= \frac{22}{7} \times \left(\frac{7}{2}\right)^2 \times 4$$

$$= \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 4$$

$$= 11 \times 7 \times 2$$

$$= 154$$

$\therefore$ Amount of soup the hospital has to prepare daily to serve 250 patients

= Volume of soup in one bowl $\times$ 250

$= 154 \times 250$

$= 38500 \text{ cm}^3$

Solution 3: Radius of each pillar, $r = 20 \text{ cm} = \frac{20}{100} \text{m} = 0.2 \text{ m}$

Height of each pillar, $h = 10 \text{ m}$

Volume of concrete mixture used in each pillar $= \pi r^2 h = \frac{22}{7} \times (0.2)^2 \times 10 \ m^3$

$\therefore$ Amount of concrete mixture required to build 14 such pillars

= Volume of concrete mixture used in each pillar $\times$ 14

$= \frac{22}{7} \times (0.2)^2 \times 10 \times 14$

$= 22 \times 0.04 \times 10 \times 2$

$= 17.6 \text{ m}^3$

Solution 4:

(i) Length of tin can, $l = 5$ cm

Breadth of tin can, $b = 4$ cm

Height of tin can, $h = 15$ cm

$\therefore$ Volume of soft drink in tin can $= l \times b \times h = 5 \times 4 \times 15 = 300 \text{ cm}^3$

(ii) Radius of plastic cylinder, $r = \frac{7}{2}$ cm

Height of plastic cylinder, $h = 10$ cm

$\therefore$ Volume of soft drink in plastic cylinder $= \pi r^2 h$

$$= \frac{22}{7} \times \left(\frac{7}{2}\right)^2 \times 10$$

$$= 11 \times 7 \times 5$$

$$= 385 \text{ cm}^3$$

So, the capacity of the plastic cylinder pack is greater than the capacity of the tin can pack.

Difference in the capacities of the two packs $= 385 - 300 = 85 \text{ cm}^3$

Thus, the capacity of the plastic cylinder pack is 85 cm^3 more than the capacity of the tin can pack.

Solution 5: Radius of each pillar, $r = \frac{50}{2} = 25 \text{ cm} = \frac{25}{100} \text{ m} = \frac{1}{4} \text{ m}$

Height of each pillar, $h = 4 \text{ m}$

$\therefore$ Surface area of each pillar $= 2\pi rh$

$$= 2 \times \frac{22}{7} \times \frac{1}{4} \times 4$$

$$= 2 \times \frac{22}{7}$$

$$= \frac{44}{7} \text{m}^2$$

Surface area of 20 pillars = Surface area of each pillar $\times$ 20

$$= \frac{44}{7} \times 20 \text{ m}^2$$

$$= \frac{880}{7} \text{ m}^2$$

Cost of cleaning = ₹ 14 per m^2

$\therefore$ Total cost of cleaning the 20 pillars

= Surface area of 20 pillars $\times$ Rate of cleaning

$= \frac{880}{7} \times 14$

$= 880 \times 2$

$= ₹ 1760$

Solution 6: Radius of the cylinder, r = 0.7 m

Curved surface area of cylinder = 4.4 m^2

(i) Let the height of the cylinder be h m.

$\therefore 2\pi rh = 4.4$

$\Rightarrow 2 \times \frac{22}{7} \times 0.7 \times h = 4.4$

$\Rightarrow 2 \times \frac{22}{7} \times \frac{7}{10} \times h = 4.4$

$\Rightarrow 44 \times \frac{1}{10} \times h = 4.4$

$\Rightarrow 4.4 \times h = 4.4$

$\Rightarrow h = \frac{4.4}{4.4}$

$\Rightarrow h = 1$ m

Thus, the height of the cylinder is 1 m.

(ii) Volume of the cylinder $= \pi r^2 h$

$= \frac{22}{7} \times (0.7)^2 \times 1$

$= \frac{22}{7} \times \left(\frac{7}{10}\right)^2$

$= \frac{22}{7} \times \frac{7}{10} \times \frac{7}{10}$

$= 22 \times \frac{1}{10} \times \frac{7}{10}$

$= \frac{154}{100}$

$= 1.54\ m^3$

Thus, the volume of the cylinder is 1.54 m^3.

Solution 7: Height of the cylinder, $h = 5$ cm

Lateral (or curved) surface area of cylinder = 94.2 cm^2

(i) Let the radius of the cylinder be r cm.

$\therefore 2\pi rh = 94.2\ cm^2$

$\Rightarrow 2 \times 3.14 \times r \times 5 = 94.2$

$$\Rightarrow r = \frac{94.2}{2 \times 3.14 \times 5}$$

$$= \frac{942 \times 100}{2 \times 314 \times 5 \times 10}$$

$$\Rightarrow r = \frac{942}{314}$$

$\Rightarrow r = 3$ cm

Thus, the radius of the cylinder is 3 cm.

(ii) Volume of the cylinder $= \pi r^2 h$

$= 3.14 \times (3)^2 \times 5$

$= 141.3\ cm^3$

Thus, the volume of the cylinder is 141.3 cm^3.

Solution 8: Height of the cylinder, $h = 1$ m $= 100$ cm

Capacity of the cylinder $= 15.4$ L $= 15.4 \times 1000 = 15400\ cm^3$

$\therefore \pi r^2 h = 15400\ cm^3$

$$\Rightarrow \frac{22}{7} \times r^2 \times 100 = 15400$$

$$\Rightarrow r^2 = \frac{15400 \times 7}{22 \times 100}$$

$$\Rightarrow r^2 = \frac{154 \times 7}{22}$$

$\Rightarrow r^2 = 7 \times 7$

$\Rightarrow r^2 = 49$

$\Rightarrow r = \sqrt{49}$

$\Rightarrow r = 7\ cm$

$\therefore$ Area of the metal sheet needed to make the cylinder

= Total surface area of the cylinder

$= 2\pi rh(r + h)$

$= 2 \times \frac{22}{7} \times 7 \times 1 \times (100 + 7)$

$= 2 \times 22 \times 107$

$= 4708 \text{ cm}^2$

Thus, the area of the metal sheet needed to make the cylinder is 0.4708 m^2.

Solution 9: Inner radius of the wooden pipe, $r = 24/2 = 12$ cm
Outer radius of the wooden pipe, $R = 28/2 = 14$ cm
Length of the wooden pipe, $h = 35$ cm
$\therefore$ Volume of wood in the pipe $= \pi(R^2 - r^2)h$

$= \frac{22}{7} \times (14^2 - 12^2) \times 35 \text{ cm}^3$

$= [22 \times (196 - 144) \times 5] \text{ cm}^3$

$= [22 \times 52 \times 5] \text{ cm}^3$

$= 5720 \text{ cm}^3$

It is given that 1 cm^3 of wood has a mass of 0.6 g.

$\therefore$ Mass of the pipe = Volume of wood in the pipe $\times$ 0.6

$= 5720 \times 0.6$

$= 3432 \text{ g}$

$= \frac{3.432}{1000} \text{ kg}$

$= 3.432$

Thus, the mass of the pipe is 3.432 kg.

Solution 10: Length of the cylindrical pipe, $h = 28 \text{ m} = 2800$ cm
Radius of the cylindrical pipe, $r = 5/2 = 2.5$ cm

$\therefore$ Total radiating surface in the system

= Curved surface area of the cylindrical pipe

$= 2\pi rh$

$= \left(2 \times \frac{22}{7} \times 2.5 \times 2800\right) \text{ cm}^2$

$= \left(2 \times 22 \times \frac{25}{10} \times 400\right) \text{ cm}^2$

$= (2 \times 22 \times 25 \times 40)\ cm^2$

$= 44000\ cm^2$

$= 4.4\ m^2$

Thus, the total radiating surface in the system is $4.4\ m^2$.

Solution 11: Here, $r = 10.5$ cm; $h = 60$ cm

Now, volume of the cylinder $= \pi r^2 h$

$= \frac{22}{7} \times (10.5)^2 \times 60\ cm^3$

$= \frac{22}{7} \times 10.5 \times 10.5 \times 60\ cm^3$

$= 22 \times 10.5 \times 1.5 \times 60\ cm^3$

$= 20790\ cm^3$

$\therefore$ Weight of cylinder = volume of cylinder $\times$ weight of cylinder per gram

$= 20790 \times 5\ g$

$= 103950\ g$

$= 103.95\ kg$

Solution 12: Curved surface area $= 1210\ cm^2$

Suppose that the height of cylinder is h cm.

We have $r = 10$ cm

Now,

$1210 = 2\pi rh$

$\Rightarrow 1210 = 2 \times \frac{22}{7} \times 10 \times h$

$\Rightarrow h = \frac{1210 \times 7}{2 \times 22 \times 10}$

$\Rightarrow h = \frac{11 \times 7}{2 \times 2}$

$\Rightarrow h = 19.25\ cm$

$\therefore$ Volume of the cylinder $= \pi r^2 h$

$= \frac{22}{7} \times 10^2 \times 19.25\ cm^3$

$= 2200 \times 2.75\ cm^3$

$= 6050\ \text{cm}^3$

Solution 13: Let r be the radius and h be the height of the cylinder.

Circumference of its base(circle) $= 110$ cm.

$\Rightarrow 2\pi r = 110$

$\Rightarrow r = \dfrac{110}{2\pi}$

$\Rightarrow r = \dfrac{110}{2 \times \frac{22}{7}}$

$\Rightarrow r = \dfrac{110 \times 7}{2 \times 22}$

$\Rightarrow r = \dfrac{35}{2}\ cm$

Curved surface area of a cylinder $= 4400\ \text{cm}^2$.

$\Rightarrow 2\pi r h = 4400$

$\Rightarrow h = \dfrac{4400}{2\pi r}$

$\Rightarrow h = \dfrac{4400}{2 \times \frac{22}{7} \times \frac{35}{2}}$

$\Rightarrow h = \dfrac{4400 \times 7 \times 2}{2 \times 22 \times 35}$

$\Rightarrow h = 40\ cm$

Also, Volume of the cylinder $= \pi r^2 h$

$= \dfrac{22}{7} \times \left(\dfrac{35}{2}\right)^2 \times 40$

$= \dfrac{22 \times 35 \times 35 \times 40}{7 \times 2 \times 2}$

$= 38500\ \text{cm}^3$

Solution 14: Suppose that the radius of the base and the height of the cylinder are $2x$cm and $3x$cm, respectively.

Then

$1617 = \pi r^2 h$

$\Rightarrow 1617 = \frac{22}{7} \times (2x)^2 \times 3x$

$\Rightarrow 1617 = \frac{22}{7} \times 4x^2 \times 3x$

$\Rightarrow 1617 = \frac{22}{7} \times 12x^3$

$\Rightarrow x^3 = \frac{1617 \times 7}{22 \times 12}$

$\Rightarrow x = \sqrt[3]{42.875}$

$\Rightarrow x = 3.5$ cm

Hence, radius $= 7$ cm; height of the cylinder $= 10.5$ cm

$\therefore$ Total surface area of the cylinder $= 2\pi rh + 2\pi r^2$

$= 2\pi r(h + r)$

$= 2 \times \frac{22}{7} \times 7(10.5 + 7)$

$= 44(17.5)$

$= 770\ \text{cm}^2$

Solution 15: Total surface area $= 462\ \text{cm}^2$

Given:

Curved surface area $= \frac{1}{3} \times$ total surface area $= \frac{1}{3} \times 462 = 154\ \text{cm}^2$

Now, total surface area $-$ curved surface area $= 2\pi rh + 2\pi r^2 - 2\pi rh$

$\Rightarrow 462 - 154 = 2\pi r^2$

$\Rightarrow 308 = 2 \times \frac{22}{7} \times r^2$

$\Rightarrow r^2 = \frac{308 \times 7}{44}$

$\Rightarrow r^2 = 49$

$\Rightarrow r = 7$ cm

Now, curved surface area $= 154\ \text{cm}^2$

$\Rightarrow 2\pi rh = 154$

$\Rightarrow 2 \times \frac{22}{7} \times 7 \times h = 154$

$\Rightarrow 2 \times 22 \times h = 154$

$\Rightarrow h = \frac{154}{2 \times 22}$

$\Rightarrow h = \frac{154}{2 \times 22}$

$\Rightarrow h = \frac{7}{2}$

$\Rightarrow h = 3.5 \text{ cm}$

$\therefore$ Volume of the cylinder $= \pi r^2 h$

$= \frac{22}{7} \times 7^2 \times 3.5 \text{ cm}^3$

$= \frac{22}{7} \times 49 \times \frac{35}{10} \text{ cm}^3$

$= 22 \times 7 \times \frac{7}{2} \text{ cm}^3$

$= 11 \times 7 \times 7 \text{ cm}^3$

$= 539 \text{ cm}^3$

Solution 16: Curved surface area $= \frac{2}{3} \times$ total surface area

$= \frac{2}{3} \times 231 \text{ cm}^2$

$= 2 \times 77 \text{ cm}^2$

$= 154 \text{ cm}^2$

Now, total surface area $-$ curved surface area $= 2\pi rh + 2\pi r^2 - 2\pi rh$

Then, $231 - 154 = 2\pi r^2$

$\Rightarrow 77 = 2 \times \frac{22}{7} \times r^2$

$\Rightarrow 77 = \frac{44}{7} \times r^2$

$\Rightarrow r^2 = 77 \times \frac{7}{44}$

$\Rightarrow r^2 = 7 \times \frac{7}{4}$

$\Rightarrow r^2 = \frac{49}{4}$

$\Rightarrow r = \frac{7}{2}$

$\Rightarrow r = 3.5 \text{ cm}$

Also, curved surface area $= 154 \text{ cm}^2$

$\Rightarrow 2\pi rh = 154$

$\Rightarrow 2 \times \frac{22}{7} \times 3.5 \times h = 154$

$\Rightarrow 2 \times \frac{22}{7} \times \frac{35}{10} \times h = 154$

$\Rightarrow 2 \times 22 \times \frac{5}{10} \times h = 154$

$\Rightarrow 2 \times 22 \times \frac{1}{2} \times h = 154$

$\Rightarrow 22 \times h = 154$

$\Rightarrow h = \frac{154}{22}$

$\Rightarrow h = 7$

$\therefore$ Volume of the cylinder $= \pi r^2 h$

$= \frac{22}{7} \times (3.5)^2 \times 7$

$= 269.5 \text{ cm}^3$

Solution 17: Suppose that the curved surface area and the total surface area of the right circular cylinder are x cm^2 and 2x cm^2.

Then we have:

$2x = 616$

$\Rightarrow x = 308 \ cm^2$

Hence, the curved surface area of the cylinder is 308 cm^2.

Let, r cm and h cm be the radius and height of the cylinder, respectively.

Then $2\pi rh + 2\pi r^2 = 616 \text{ cm}^2$ and $2\pi rh = 308 \text{ cm}^2$

$\therefore 2\pi rh + 2\pi r^2 - 2\pi rh = 616 - 308$

$\Rightarrow 2\pi r^2 = 308$

$$\Rightarrow 2 \times \frac{22}{7} \times r^2 = 308$$

$$\Rightarrow r^2 = 308 \times \frac{7}{44}$$

$$\Rightarrow r^2 = 7 \times 7$$

$$\Rightarrow r^2 = 49$$

$$\Rightarrow r = 7 \text{ cm}$$

Now, $2\pi rh = 308 \text{ cm}^2$

$$\Rightarrow 2 \times \frac{22}{7} \times 7 \times h = 308$$

$$\Rightarrow h = \frac{308}{44}$$

$$\Rightarrow h = 7 \text{ cm}$$

$\therefore$ Volume of the cylinder $= \pi r^2 h$ cubic cm

$$= \frac{22}{7} \times 7^2 \times 7 \text{ cm}^3$$

$$= 22 \times 49 \text{ cm}^3$$

$$= 1078 \text{ cm}^3$$

Solution 18: Given: Diameter of the cylindrical bucket $= 28$ cm

i.e., radius $= 14$ cm

Height of the cylindrical bucket, $h_1 = 72$ cm

Length of the rectangular tank, $l = 66$ cm

Breadth of the rectangular tank, $b = 28$ cm

Let the height of the rectangular tank be h cm.

The water from the cylindrical bucket is emptied into the rectangular tank.

i.e., volume of the bucket = volume of the tank

$$\Rightarrow \pi r^2 h_1 = l \times b \times h$$

$$\Rightarrow \frac{22}{7} \times 14^2 \times 72 = 66 \times 28 \times h$$

$$\Rightarrow \frac{22}{7} \times 14 \times 14 \times 72 = 66 \times 28 \times h$$

$\Rightarrow 22 \times 2 \times 14 \times 72 = 66 \times 28 \times h$

$\Rightarrow h = \dfrac{22 \times 2 \times 14 \times 72}{66 \times 28}$

$\Rightarrow h = \dfrac{14 \times 72}{3 \times 14}$

$\Rightarrow h = 24 \text{ cm}$

$\therefore$ Height of the rectangular tank $= 24$ cm

Solution 19: Height of the barrel $= h = 7$ cm

Radius of the barrel $= r = 2.5$ mm $= 0.25$ cm

Volume of the barrel $= \pi r^2 h$

$$= \frac{22}{7} \times (0.25)^2 \times 7$$

$$= 22 \times 0.25 \times 0.25$$

$$= 1.375 \text{ cm}^3$$

i.e., 1.375 cm^3 of ink is used for writing 330 words

Now, number of words that could be written with one-fifth of a litre,

$$i.e., \frac{1}{5} \times 1000\ cm^3 = 330 \times \frac{1}{1.375} \times \frac{1}{5} \times 1000 = 48000$$

$\therefore$ 48000 words would use up a bottle of ink containing one-fifth of a litre.

Solution 20: Let r cm be the radius of the wire and h cm be the height of the wire.

Volume of the gold $= 1 \text{ cm}^3$

1 cm^3 of gold is drawn into a wire of diameter 0.1 mm.

Here, $r = \dfrac{0.1}{2} \text{mm} = \dfrac{1}{20} \text{mm} = \dfrac{1}{200}\ cm$

$\therefore\ \pi r^2 h = 1$

$$\Rightarrow \frac{22}{7} \times \left(\frac{1}{200}\right)^2 \times h\ = 1$$

$$\Rightarrow h = \frac{40000 \times 7}{22}$$

$$\Rightarrow h = 12727.27\ cm$$

$$\Rightarrow h = 127.27\ m$$

Hence, length of the wire $= 127.27$ m

Solution 21: Internal radius of the pipe $= 1.5$ cm

External radius of the pipe $= (1.5 + 1)$ cm $= 2.5$ cm

Height of the pipe $= 1$ m $= 100$ cm

Volume of the cast iron = total volume of the pipe − internal volume of the pipe

$$= \pi \times (2.5)^2 \times 100 - \pi \times (1.5)^2 \times 100$$

$$= \pi \times 100 \times [(2.5)^2 - (1.5)^2]$$

$$= \frac{22}{7} \times 100 \times [6.25 - 2.25]$$

$$= \frac{2200}{7} \times 4$$

$$= \frac{8800}{7} cm^3$$

1 cm^3 of cast iron weighs 21 g.

$\therefore$ Weight of $\frac{8800}{7}$ cm^3 cast iron

$$= \frac{8800}{7} \times \frac{21}{1000}\ kg$$

$$= 88 \times \frac{3}{10}\ kg$$

$$= \frac{264}{10}\ kg$$

$$= 26.4\ kg$$

Solution 22: Inner radius of the cylindrical tube, $r = 5.2$ cm

Height of the cylindrical tube, $h = 25$ cm

Outer radius of the cylindrical tube, $R = (5.2 + 0.8)$ cm $= 6$ cm

$\therefore$ Volume of the metal = external volume − internal volume

$= \pi R^2 h - \pi r^2 h$ (where R and r are the outer and inner radii, respectively)

$$= \frac{22}{7} \times 25 \times (6^2 - 5.2^2)$$

$$= \frac{22}{7} \times 25 \times (36 - 27.04)$$

$= \frac{22}{7} \times 25 \times 8.96$

$= 704 \text{ cm}^3$

Solution 23: Height of the cylindrical tank, $h = 1$ m

Radius of the cylindrical tank, $r = \frac{140}{2} = 70 \text{ cm} = \frac{70}{100} \text{m} = 0.7 \text{ m}$

∴ Area of the metal sheet required to make the cylindrical tank

= Total surface area of the cylindrical tank

$= 2\pi r(r + h)$

$= 2 \times \frac{22}{7} \times 0.7 \times (0.7 + 1)$

$= 2 \times \frac{22}{7} \times 0.7 \times 1.7$

$= 7.48 \text{ m}^2$

Thus, the area of metal sheet required to make the cylindrical tank is 7.48 m^2.

Solution 24: Radius of the vessel, $R = 15$ cm

Height of orange juice in the vessel, $H = 32$ cm

∴ Volume of orange juice in the vessel $= \pi R^2 H$

$= \pi \times (15)^2 \times 32 \text{ cm}^3$

Radius of the glass, $r = 3$ cm

Height of orange juice in the glass, $h = 8$ cm

∴ Volume of orange juice in each glass $= \pi r_2 h$

$= \pi \times (3)^2 \times 8 \text{ cm}^3$

Number of glasses of orange juice sold by the juice seller

= Volume of orange juice in the vasselVolume of orange juice in each glass

$= \pi \times (15)^2 \times 32\pi \times (3)^2 \times 8$

$= 100$

Rate of each glass of orange juice = ₹ 15

∴ Total money received by the juice seller

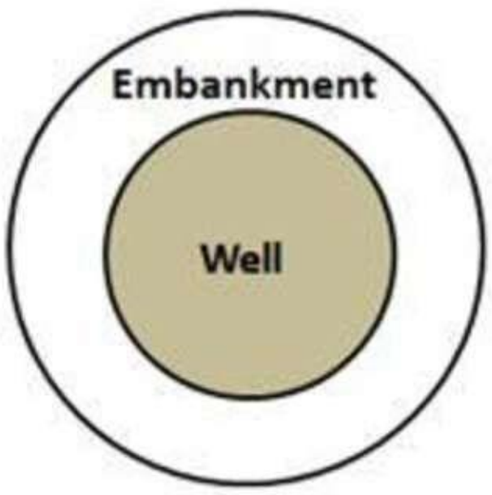
= Number of glasses of orange juice sold by the juice seller × Rate of each glass of orange juice

$= 100 \times 15$

= ₹ 1,500

Thus, the total money received by the juice seller by selling the juice completely is ₹ 1,500.

Solution 25: Inner radius of the well, $r = 10/2 = 5$ m
Depth of the well, $h = 8.4$ m



Suppose the outer radius of the embankment is R m.
Width of the embankment = 7.5 m

$\therefore R - r = 7.5$ m

$\Rightarrow R = 7.5 + 5$

$\Rightarrow R = 12.5$ m

Let the height of the embankment be H m.
Now,
Volume of earth used to form the embankment = Volume of earth dugged out of the well

$\therefore \pi(R^2 - r^2)H = \pi r^2 h$

$\Rightarrow (R^2 - r^2)H = r^2 h$

$\Rightarrow H = \dfrac{r^2 h}{R^2 - r^2}$

$\Rightarrow H = \dfrac{5^2 \times 8.4}{(12.5)^2 - (5)^2}$

$\Rightarrow H = \dfrac{25 \times 8.4}{(12.5 + 5)(12.5 - 5)}$ $[\because a^2 - b^2 = (a + b)(a - b)]$

$\Rightarrow H = \dfrac{25 \times 8.4}{17.5 \times 7.5}$

$\Rightarrow H = \dfrac{25 \times 84 \times 10 \times 10}{175 \times 75 \times 10}$

$$\Rightarrow H = \frac{84 \times 10}{175 \times 3}$$

$$\Rightarrow H = \frac{28 \times 2}{35}$$

$$\Rightarrow H = \frac{28 \times 2}{35}$$

$$\Rightarrow H = \frac{56}{35}$$

$\Rightarrow H = 1.6$ m

Thus, the height of the embankment is 1.6 m.

Solution 26: Speed of the water $= 30$ cm/s

Area of the cross section $= 5$ cm^2

Volume of the water flowing out of the pipe in one second

$=$ Area of the cross section $\times$ 30 cm

$= 5 \times 30$

$= 150$ cm^3

Now, 1 minute $= 60$ seconds

$\therefore$ Volume of the water flowing out of the pipe in 60 seconds

$=$ Volume of the water flowing out of the pipe in one second $\times$ 60

$= 150 \times 60$

$= 9000$ cm^3

$= 9000/1000$ ($\because$ 1 L $= 1000$ cm^3)

$= 9$ L

Thus, 9 L of water flows out of the given pipe in 1 minute.

Solution 27: Radius of the water tank, $R = 1.421.42 = 0.7$ m

Height of the water tank, $H = 2.1$ m

$\therefore$ Capacity of the water tank $= \pi R^2 H = \pi(0.7)^2 \times 2.1$ m^3

Speed of the water flow $= 2$ m/s

Radius of the pipe, $r = 3.5/2 = 1.75$ cm $= 0.0175$ m

Area of the cross section of the pipe $= \pi r^2 = \pi\,(0.0175)^2$

Volume of the water flowing out of the pipe in one second

= Area of the cross section of the pipe $\times$ 2 m

$= \pi(0.0175)^2 \times 2$ m^3

Let the time taken to fill the tank be t seconds.

$\therefore$ Volume of the water flowing out of the pipe in t seconds

= Volume of the water flowing out of the pipe in one second $\times\ t$

$= \pi\,(0.0175)^2 \times 2 \times$ t m^3

Now,

Volume of the water flowing out of the pipe in t seconds = Capacity of the water tank

$\therefore \pi(0.0175)^2 \times 2 \times t = \pi(0.7)^2 \times 2.1$

$$\Rightarrow t = \frac{(0.7)^2 \times 2.1}{(0.0175)^2 \times 2}$$

$$\Rightarrow t = \frac{1680}{60}\ s$$

$\Rightarrow t = 28$ minutes

Thus, the tank will be filled in 28 minutes.

Solution 28: Volume of the rectangular solid of iron $= 32$ cm $\times$ 22 cm $\times$ 14 cm

Radius of the container, $r = 56/2 = 28$ cm

Let the rise in the level of water in the container when rectangular solid of iron is submerged in it be h cm.

$\therefore$ Volume of the water displaced in the container $= \pi r^2 h = \pi \times (28)^2 \times$ h

When the rectangular solid of iron is submerged in the container, then the volume of water displaced in the container is equal to the volume of the rectangular solid of iron.

$$\therefore \frac{22}{7} \times (28)^2 \times h = 32 \times 22 \times 14$$

$$\Rightarrow h = \frac{32 \times 22 \times 14 \times 7}{22 \times (28)^2}$$

$\Rightarrow h = 4\ cm$

Thus, the rise in the level of water in the container is 4 cm.

Solution 29: Height of the tube-well, $h = 280$ m

Radius of the tubewell, $r = \frac{3}{2}$ m

$\therefore$ Volume of the tubewell $= \pi r^2 h = \frac{22}{7} \times \left(\frac{3}{2}\right)^2 \times 280 = 1980 \text{ m}^3$

Rate of sinking the tube-well = ₹ 15 per cubic metre

$\therefore$ Cost of sinking the tube-well

= Volume of the tube-well × Rate of sinking the tube-well

$= 1980 \times 15$

= ₹ 29,700

Thus, the cost of sinking the tube-well is ₹ 29,700.

Inner curved surface of the tube-well $= 2\pi rh$

$= 2 \times \frac{22}{7} \times \frac{3}{2} \times 280$

$= 22 \times 3 \times 40$

$= 2640 \text{ m}^2$

Rate of cementing = ₹ 10 per square metre

$\therefore$ Cost of cementing the inner curved surface of the tube-well

= Inner curved surface of the tube-well × Rate of cementing

$= 2640 \times 10$

= ₹ 26,400

Thus, the cost of cementing the inner curved surface of the tube-well is ₹ 26,400.

Solution 30: Mass of copper wire $= 13.2 \text{ kg} = 13.2 \times 1000 = 13200 \text{ g}$ ($\because 1 \text{ kg} = 1000$ g)

Volume of 8.4 g of copper wire $= 1 \text{ cm}^3$

$\therefore$ Volume of 13200 g (or 13.2 kg) of copper wire $= \frac{13200}{8.4} \text{ cm}^3$

Let the length of the copper wire be l cm.

Radius of the copper wire, $r = \frac{4}{2} = 2 \text{ mm} = \frac{2}{10} \text{cm} = 0.2 \text{ cm}$

$\therefore$ Volume of the copper wire $= \pi r^2 l$

$$\Rightarrow \frac{22}{7} \times (0.2)^2 \times l = \frac{13200}{8.4}$$

$$\Rightarrow l = \frac{13200 \times 7}{22 \times (0.2)^2 \times 8.4}$$

$$\Rightarrow l = 12500 \text{ cm}$$

$$\Rightarrow l = \frac{12500}{100} \text{m}$$

$$\Rightarrow l = 125\, m$$

Thus, the length of the copper wire is 125 m.

Solution 31:

Total cost of paining the inner curved surface of the cylindrical vessel = ₹ 3,300
Rate of painting = ₹ 30 per m^2

(i) Inner curved surface area of the vessel

$$= \frac{\text{Total cost of painting the inner curved surface of the cylindrical vassel}}{\text{Rate of painting}}$$

$$= \frac{3300}{30}$$

$=110 \text{ m}^2$

Thus, the inner curved surface area of the vessel is 110 m^2.

(ii) Depth of the vessel, $h = 10$ m
Let the inner radius of the base be r m.
$\therefore$ Inner curved surface area of the vessel $= 2\pi rh$

$$\Rightarrow 110 = 2 \times \frac{22}{7} \times r \times 10$$

$$\Rightarrow r = \frac{110 \times 7}{2 \times 22 \times 10}$$

$$\Rightarrow r = 1.75\, m$$

Thus, the inner radius of the base is 1.75 m.

(iii) Capacity of the vessel $= \pi r^2 h$

$$= \frac{22}{7} \times (1.75)^2 \times 10$$

$$= \frac{22}{7} \times \left(\frac{175}{100}\right)^2 \times 10$$

$$= 22 \times \frac{175}{100} \times \frac{1755}{100} \times 10$$

$$= 22 \times \frac{175}{100} \times \frac{25}{10}$$

$$= 22 \times \frac{175}{100} \times \frac{5}{2}$$

$$= 11 \times \frac{175}{100} \times 5$$

$$= \frac{96.25}{100}$$

$$= 96.25 \text{ m}^3$$

Thus, the capacity of the vessel is 96.25 m^3.

Solution 32: Let the inner and outer radii of the tube be r cm and R cm, respectively.

Length of the cylindrical tube, $h = 14$ cm

Given,

Outer curved surface of the cylinder − Inner curved surface of the cylinder = 88 cm^2

$$\Rightarrow 2\pi Rh - 2\pi rh = 88$$

$$\Rightarrow 2\pi(R - r)h = 88$$

$$\Rightarrow 2 \times \frac{22}{7}(R - r) \times 14 = 88$$

$$\Rightarrow R - r = \frac{88 \times 7}{2 \times 22 \times 14}$$

$$\Rightarrow R - r = 1 \qquad \ldots \text{(i)}$$

It is given that the volume of the tube = 176 cm^3

$$\Rightarrow \text{External volume - Internal volume} = 176 \text{ cm}^3$$

$$\Rightarrow \pi R^2 h - \pi r^2 h = 176$$

$$\Rightarrow \pi(R^2 - r^2)h = 176$$

$$\Rightarrow \frac{22}{7} \times (R + r)(R - r) \times 2 = 176$$

$$\Rightarrow 22 \times (R + r) \times 1 \times 2 = 176 \qquad [\because R - r = 1]$$

$$\Rightarrow 44 \times (R + r) = 176$$

$\Rightarrow R + r = \frac{176}{44}$

$\Rightarrow R + r = 4$ ... (ii)

Adding (i) and (ii), we get

$2R = 5$

$\Rightarrow R = 2.5$ cm

Putting $R = 2.5$ cm in (ii), we get

$2.5 + r = 4$

$\Rightarrow r = 4 - 2.5$

$\Rightarrow r = 1.5$ cm

Thus, the inner and outer radii of the tube are 1.5 cm and 2.5 cm, respectively.

Solution 33: The dimensions of the rectangular sheet of paper are 30 cm × 18 cm.

Let V_1 and V_2 be the volumes of the cylinders formed by rolling the rectangular sheet of paper along its length (i.e. 30 cm) and breadth (i.e. 18 cm), respectively.

Suppose r_1 and h_1 be the radius and height of the cylinder formed by rolling the rectangular sheet of paper along its length, respectively.

$\therefore 2\pi r_1 = 30$

$\Rightarrow r_1 = \frac{30}{2\pi}\ cm$

$h_1 = 18$ cm

$\therefore V_1 = \pi r_1^2 h_1$

$= \pi\left(\frac{30}{2\pi}\right)^2 \times 18\ cm^3$

$= \pi\left(\frac{15}{\pi}\right)^2 \times 18\ cm^3$

$= \pi\left(\frac{225}{\pi^2}\right) \times 18\ cm^3$

$= \left(\frac{225}{\pi} \times 18\right)\ cm^3$

Also, suppose r_2 and h_2 be the radius and height of the cylinder formed by rolling the rectangular sheet of paper along its breadth, respectively.

$\therefore 2\pi r^2 = 18$
$\Rightarrow r^2 = 182\pi\ cm$

$h_2 = 30$ cm

$\therefore V_2 = \pi r_2^2 h_2$

$= \pi\left(\frac{18}{2\pi}\right)^2 \times 30\ cm^3$

$= \pi\left(\frac{9}{\pi}\right)^2 \times 30\ cm^3$

$= \pi\left(\frac{81}{\pi^2}\right) \times 30\ cm^3$

$= \left(\frac{81}{\pi} \times 30\right)\ cm^3$

Now,

$$\frac{V_1}{V_2} = \frac{\left(\frac{225}{\pi} \times 18\right)\ cm^3}{\left(\frac{81}{\pi} \times 30\right)\ cm^3} = \frac{225 \times 18}{81 \times 30} = \frac{5}{3} = 5:3$$

$\therefore V_1 : V_2 = 5 : 3$

Thus, the ratio of the volumes of the two cylinders thus formed is 5 : 3.